

Solutions: Normal Distribution Practice Problems

1. According to the Empirical Rule, what percentage of data lies within: $\pm 1\sigma$, $\pm 2\sigma$, $\pm 3\sigma$?

Solution: According to the Empirical Rule (68-95-99.7 Rule):

- Within $\pm 1\sigma$: **68%** of the data lies here.
- Within $\pm 2\sigma$: **95%** of the data lies here.
- Within $\pm 3\sigma$: **99.7%** of the data lies here.

2. A dataset has a mean of 50 and a standard deviation of 10. According to the Empirical Rule, find the range where approximately 95% of the data lies.

Solution: Approximately 95% of the data lies within 2 standard deviations ($\pm 2\sigma$) of the mean (μ).

$$\text{Lower bound} = \mu - 2\sigma = 50 - 2(10) = 50 - 20 = 30$$

$$\text{Upper bound} = \mu + 2\sigma = 50 + 2(10) = 50 + 20 = 70$$

The range is **30 to 70**.

3. A student scored 85 in an exam where the mean score is 70 and the standard deviation is 5. Calculate the Z-score of the student.

Solution: The formula for a Z-score is $Z = (X - \mu) / \sigma$.

$$Z = (85 - 70) / 5$$

$$Z = 15 / 5 = 3$$

The student's Z-score is **3.0**.

4. Given $Q1=25$ and $Q3=45$ calculate the IQR and explain what it represents about the dataset.

Solution: The Interquartile Range (IQR) is calculated as:

$$IQR = Q3 - Q1$$

$$IQR = 45 - 25 = 20$$

Meaning: The IQR of **20** represents the "Middle 50%" of the data. It measures the spread or dispersion of the middle half of the dataset, showing where the bulk of the typical values are concentrated while ignoring outliers.

5. A dataset has a mean (μ) of 80 and a standard deviation (σ) of 5. Using the Empirical Rule, determine the range where approximately 68% of the data lies.

Solution: Approximately 68% of the data lies within 1 standard deviation ($\pm 1\sigma$) of the mean (μ).

$$\text{Lower bound} = \mu - 1\sigma = 80 - 5 = 75$$

$$\text{Upper bound} = \mu + 1\sigma = 80 + 5 = 85$$

The range is **75 to 85**.

6. The heights of students are normally distributed with a mean of 160 cm and a standard deviation of 8 cm. Find the range that contains approximately 95% of the students' heights.

Solution: Approximately 95% of the data lies within 2 standard deviations ($\pm 2\sigma$) of the mean.

$$\text{Lower bound} = 160 - 2(8) = 160 - 16 = 144 \text{ cm}$$

$$\text{Upper bound} = 160 + 2(8) = 160 + 16 = 176 \text{ cm}$$

The range containing approximately 95% of the students' heights is **144 cm to 176 cm**.

7. A student scored 72 on a test where the mean score is 60 and the standard deviation is 6. Calculate the Z-score of the student.

Solution: Using the Z-score formula $Z = (X - \mu) / \sigma$:

$$Z = (72 - 60) / 6$$

$$Z = 12 / 6 = 2$$

The student's Z-score is **2.0**.